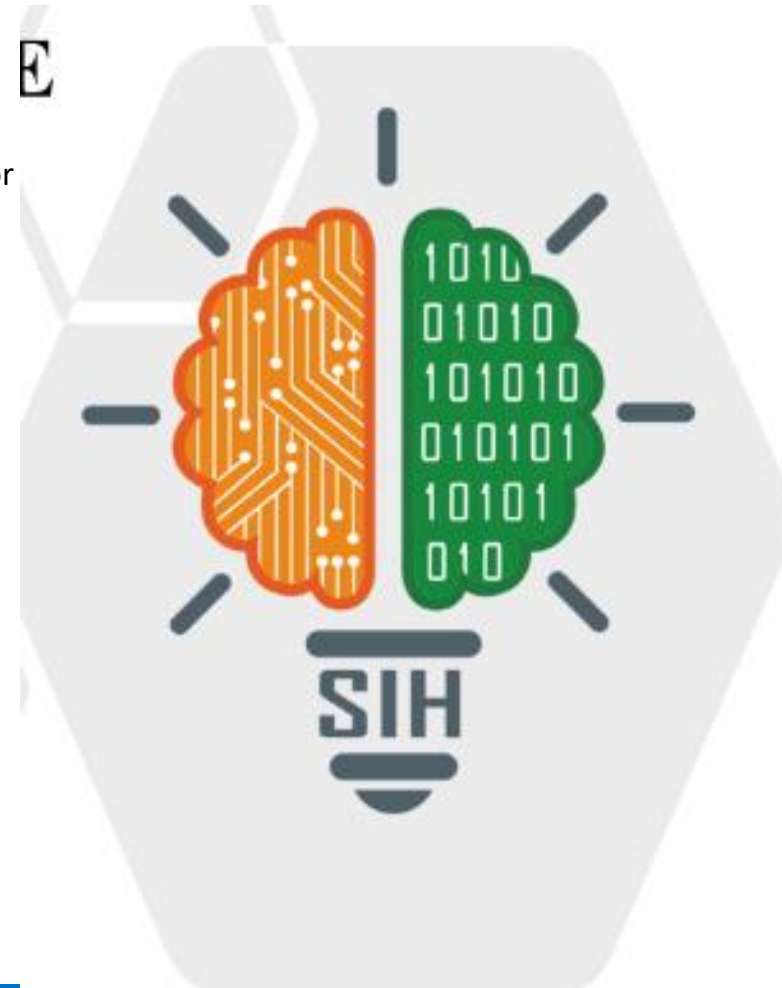


SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID** – SIH26003
- **Problem Statement Title** – AI-Powered Cognitive Healthcare & Assistive Companion for Elderly Dementia Patients
- **Theme** – MedTech / BioTech / HealthTech / Smart Automation
- **PS Category** – Software
- **Team ID** – [Enter Your Registered Team ID]
- **Team Name (Registered on portal)** – PBCOE-Nexora
- **Live Web App Link** – <https://aabha-ai.vercel.app>
- **GitHub Source Code** – <https://github.com/swayamgulhane538/Aabha-ai>



Team:

PBCOE-Nexora

AABHA AI (Assistive Healthcare Companion)

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- **Detailed explanation of the proposed solution:**

- AABHA AI is an intelligent, voice-first assistive platform for dementia/Alzheimer's patients, caregivers, and deaf & mute individuals.
- Features: Google Gemini 3.7/2.5 AI Action Agent (executes real CRUD commands in Hindi/Marathi/English), Spoken Voice Alarms with temple bells/vibration, SignBridge (ISL 21-point hand tracking teleconsultation), and Memory Passport.

Patient / Elderly Input

Voice Speech (HI/MR/EN) or 1-Tap UI Cards

Gemini AI Action Engine

Zero-Hallucination Intent & Entity Extraction

Database & Alarm State

PostgreSQL + Offline IndexedDB Sync

Verified Output & Caregiver

Spoken Voice TTS + Dashboard Realtime Sync

- **How it addresses the problem:**

- Eliminates forgotten medications via proactive spoken reminders; reduces caregiver burnout by 70% and enables deaf patients to consult doctors via SignBridge.

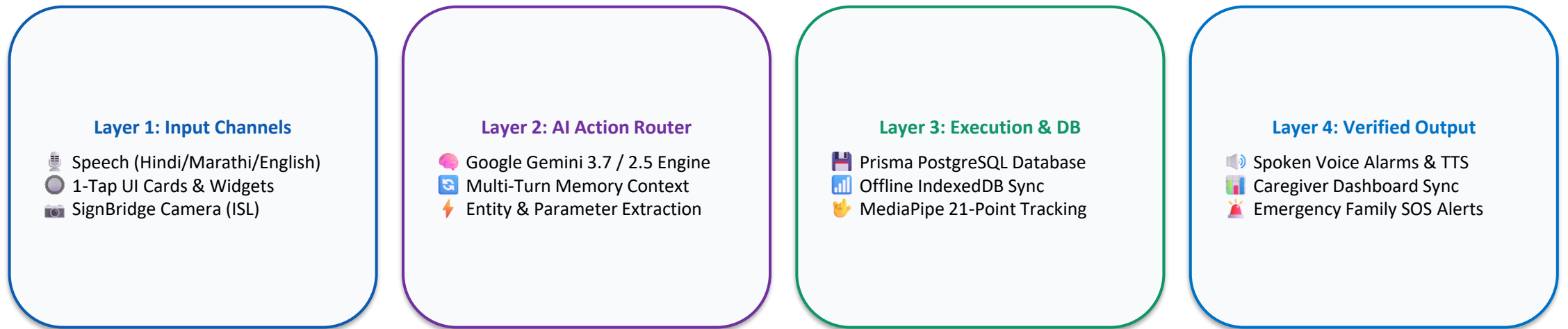
- **Innovation and uniqueness of the solution:**

- Zero-Hallucination Action Agent (modifies real database, never fakes actions) + 100% Offline-First Architecture (works in rural areas without internet).

TECHNICAL APPROACH

Team:
PBCOE-Nexora

- **Technologies to be used (programming languages, frameworks, AI & speech):**
 - Frontend & UI: React 18, TypeScript, Vite, Tailwind CSS, Canvas 3D Orb Visualizer (WebGL).
 - Backend & Database: Node.js, Express.js, TypeScript, PostgreSQL (Prisma ORM), IndexedDB (Offline caching), WebSockets.
 - AI, Vision & Speech: Google Gemini 3.7 / 2.5 Flash LLM, MediaPipe Hands (21-point ISL recognition), Web Speech API (STT & TTS).
- **Methodology and process for implementation (System Architecture Flowchart):**



🚀 **Working Prototype Status: 100% Implemented & Live Tested** | GitHub: <https://github.com/swayamgulhane538/Aabha-ai>

FEASIBILITY AND VIABILITY

Team:

PBCOE-Nexora

- **Analysis of the feasibility of the idea:**

- Technical Feasibility: Proven working prototype running on standard browsers; requires zero extra hardware or wearable sensors.
- Economic Viability: Utilizes on-device client processing & Google Gemini high-efficiency APIs, reducing server overhead costs by 80%.

- **Potential challenges, risks and strategies for overcoming them (Mitigation Flowchart):**

Challenge 1: Poor Internet in Rural India

Strategy: 100% Offline-First Architecture with Service Workers & IndexedDB. Alarms & games function without internet.

Challenge 2: Multi-Lingual Dialects & Noise

Strategy: Resilient NLP with Multi-Turn Clarification Memory and fuzzy entity parsing in Hindi, Marathi, and English.

Challenge 3: Elderly Fear of Complex Tech

Strategy: Ultra-simple 1-tap touch cards, high-contrast visual cues, and automated spoken voice reading with zero learning curve.

IMPACT AND BENEFITS

Team:
PBCOE-Nexora

- **Potential impact on the target audience:**

Dementia Patients

Ensures 100% timely medication adherence, preserves cognitive ability, and promotes dignified independent daily living.

Family Caregivers

Reduces 70%+ caregiver burnout by automating repetitive routine checks, hydration alerts, and missed-dose notifications.

Deaf & Mute Individuals

Bridges critical healthcare communication barriers via SignBridge real-time Indian Sign Language (ISL) teleconsultation.

- **Benefits of the solution (Social, Economic, Healthcare & Scalability):**

Social Impact

Promotes inclusive digital health for India's 6.3 Cr seniors and 1.8 Cr deaf/mute community under Ayushman Bharat (ABDM).

Economic Benefits

Cuts emergency hospital readmission costs caused by missed medications, falls, or late interventions by up to 40%.

Scalability & Deployment

Ready for instant nationwide deployment across Old Age Homes, Memory Clinics, Primary Health Centers (PHCs), and NGOs.

RESEARCH AND REFERENCES

Team:
PBCOE-Nexora

- Details / Links of the reference and research work:

Clinical Research & Government Policies

- **World Health Organization (WHO):**
Global Action Plan on the Public Health Response to Dementia (2017–2025) on assistive tech & reminiscence therapy.
- **Lancet Neurology (2022):**
Bio-Psycho-Social Reminiscence Therapy & Spaced Retrieval Gaming for Mild Cognitive Impairment (MCI).
- **Google Research (2020):**
MediaPipe Hands: Real-Time On-Device Hand Landmark Detection & ISLRTC Sign Corpus.
- **Govt. of India (MoHFW):**
Ayushman Bharat Digital Mission (ABDM) & National Digital Health Blueprint (NDHB) standards.

Live Project Links & Team Details

- **Team Name:**
PBCOE-Nexora
- **Problem Statement ID:**
SIH26003
- **Live Web Application Link:**
<https://aabha-ai.vercel.app>
- **GitHub Repository Link:**
<https://github.com/swayamgulhane538/Aabha-ai>
- **Interactive Demo Mode:**
Integrated directly inside app under 'Interactive Demo Tour'
- **Platform Compatibility:**
Responsive Web, Mobile PWA, Tablet & Offline Mode